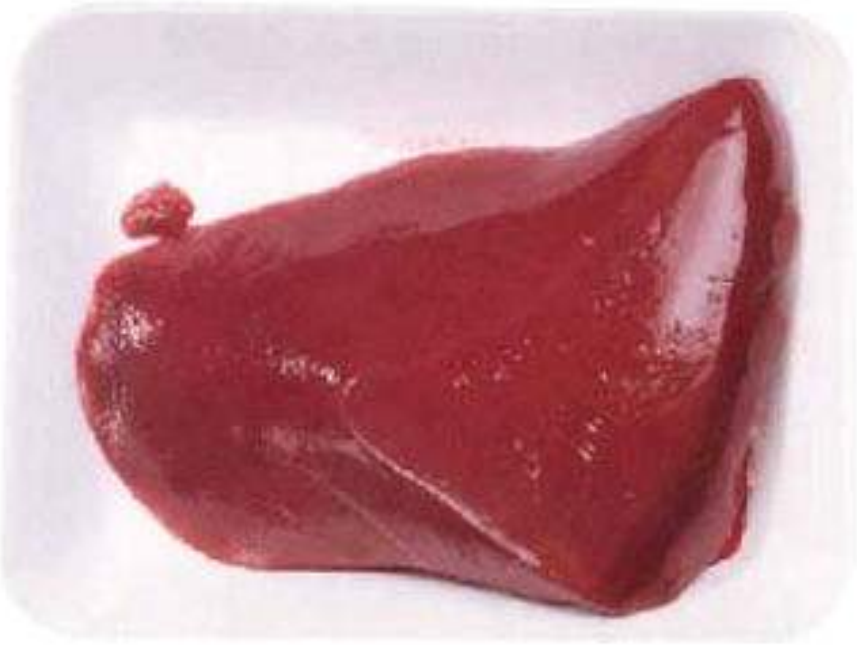
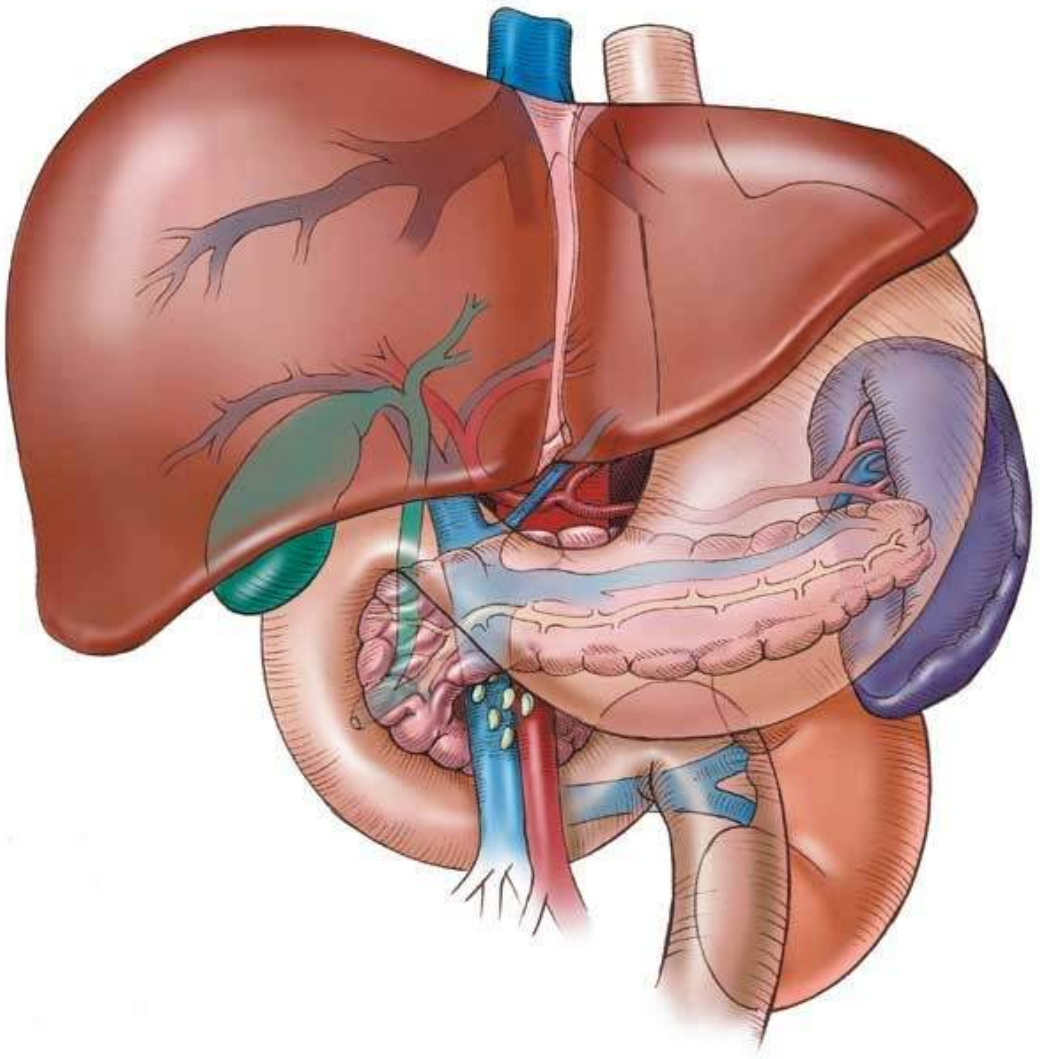


The Liver Lecture

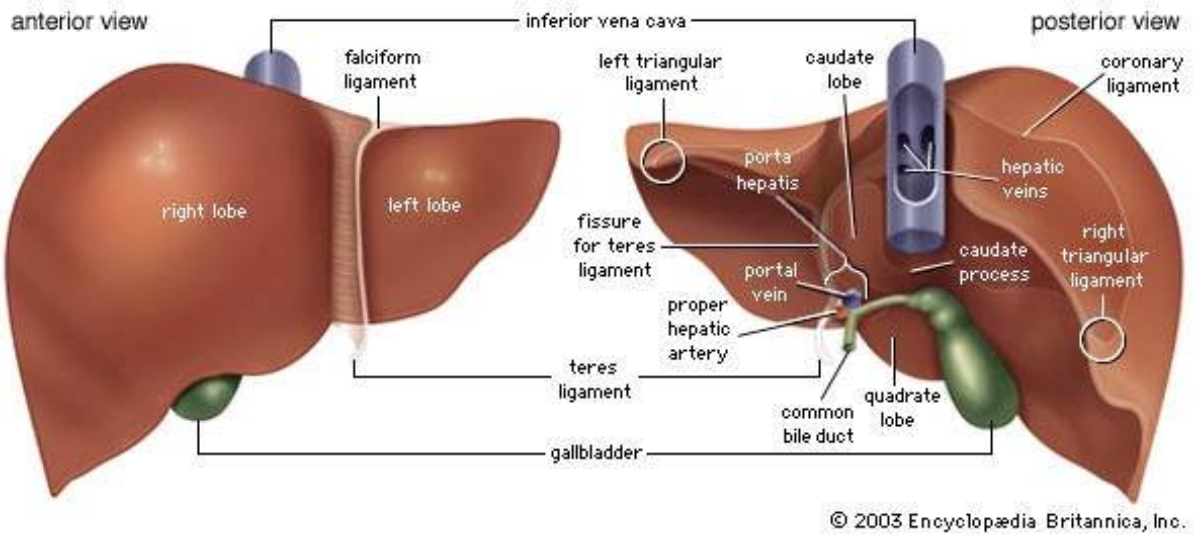
The Liver

- The largest single organ in the human body.
- In an adult, it weighs about three pounds and is roughly the size of a football.
- Located in the upper right-hand part of the abdomen, behind the lower ribs.



Gross Anatomy

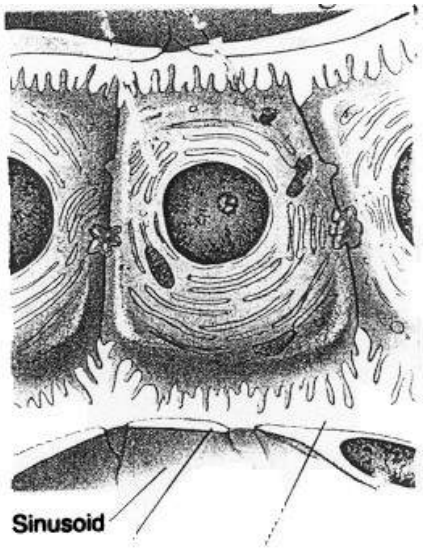
- The liver is divided) into four lobes: the **right** (the largest lobe), **left**, **quadrate** and **caudate** lobes.
- Supplied with blood via the protal vein and hepatic artery.
- Blood carried away by the hepatic vein.
- It is connected to the diaphragm and abdomainal walls by five ligaments.
- Gall Bladder
  - Muscular bag for the storage, concentration, acidification and delivery of bile to small intestine

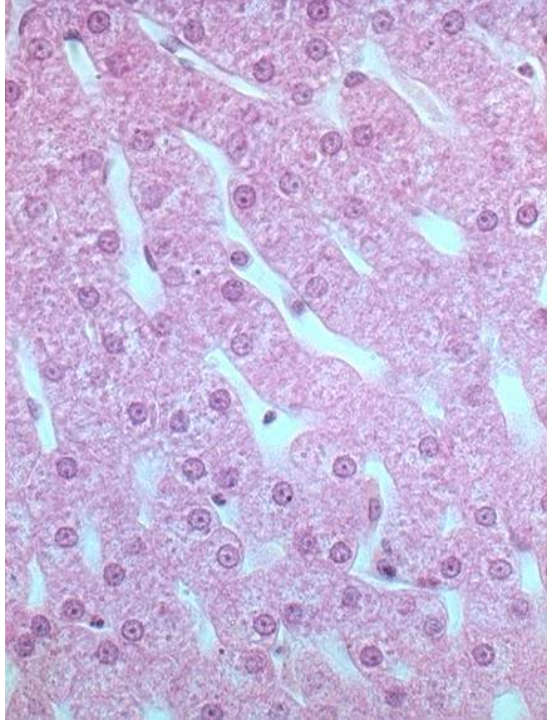


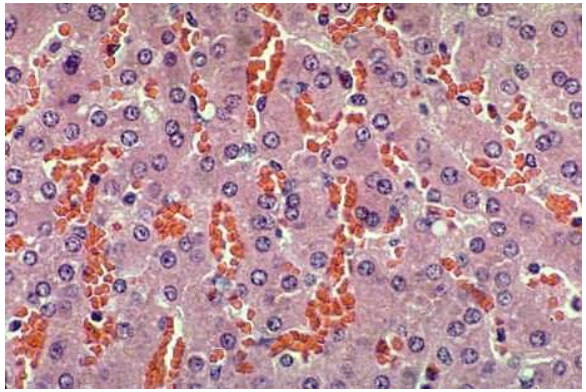
- The liver is the only human organ that has the remarkable property of self-regeneration. If a part of the liver is removed, the remaining parts can grow back to its original size and shape.

Microscopic Anatomy

- Hepatocyte—functional unit of the liver
  - Cuboidal cells
  - Arranged in plates→lobules
  - Nutrient storage and release
  - Bile production and secretion
  - Plasma protein synthesis
  - Cholesterol Synthesis



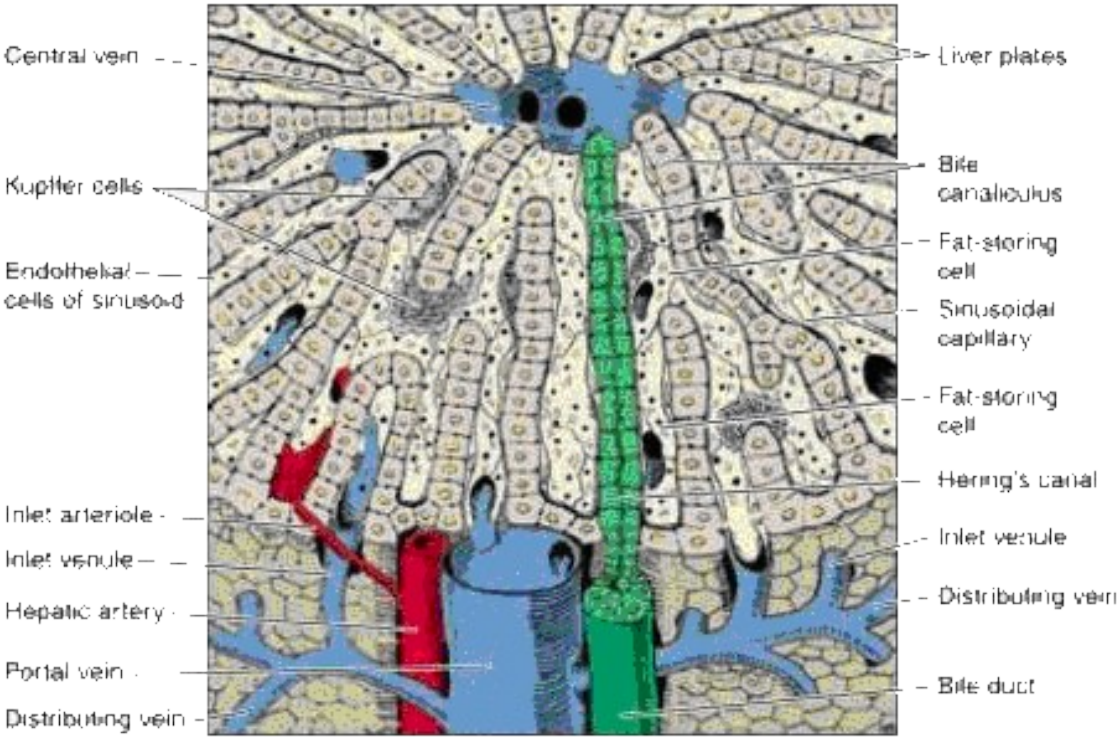




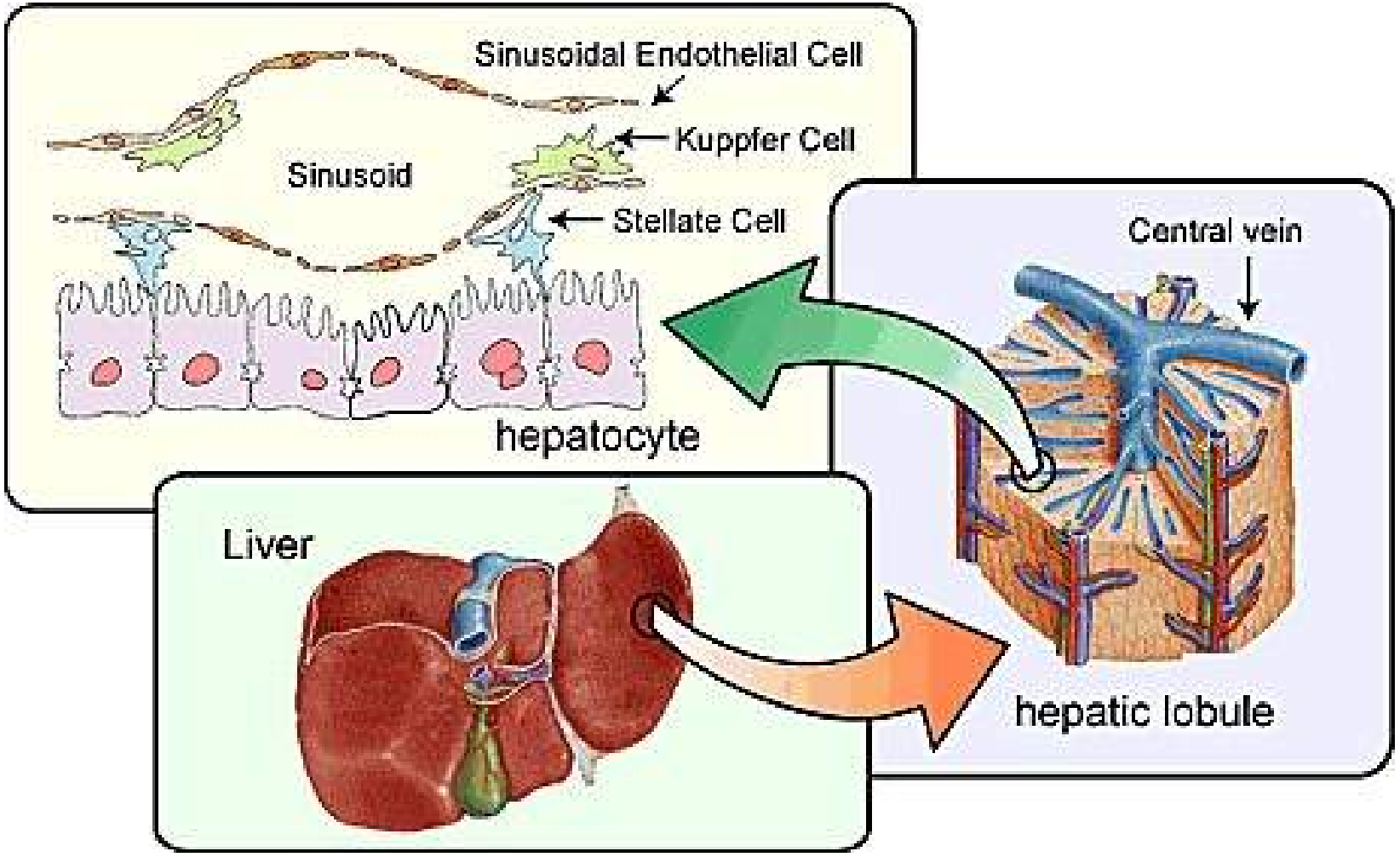


Microscopic Anatomy

- Kuppfer cells
  - Phagocytic cells
- Fat Storing Cells
- Sinusoids
  - Fenestrated vessel
  - Wider than capillaries
  - Lined with endothelial cells
  - Blood flow
- Branches of the hepatic artery
- Branches of the Hepatic portal vein, central vein
- Bile canaliculi

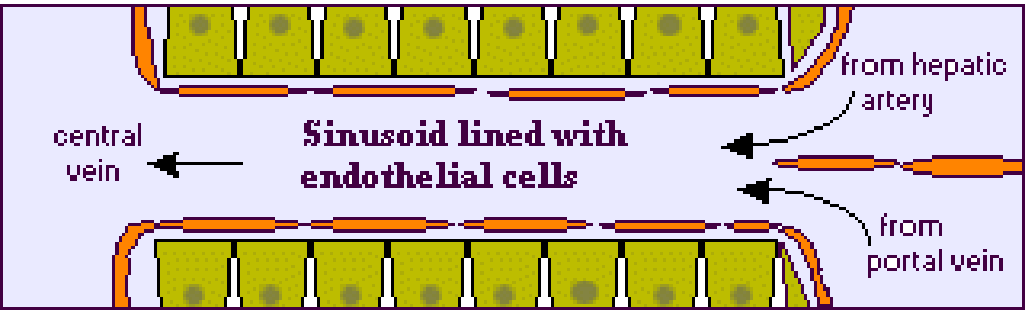


Microscopic Anatomy



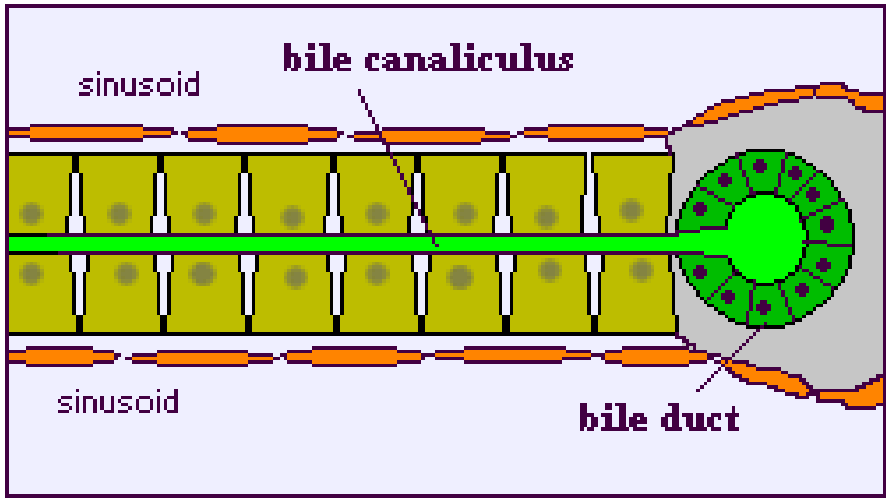
Blood and Bile Flow in Opposite Directions

- **Blood Flow**



- Deoxygenated blood from stomach or small intestine→Hepatic PortalVein→venules→sinusoids→ cental vein→hepatic vein→vena cava

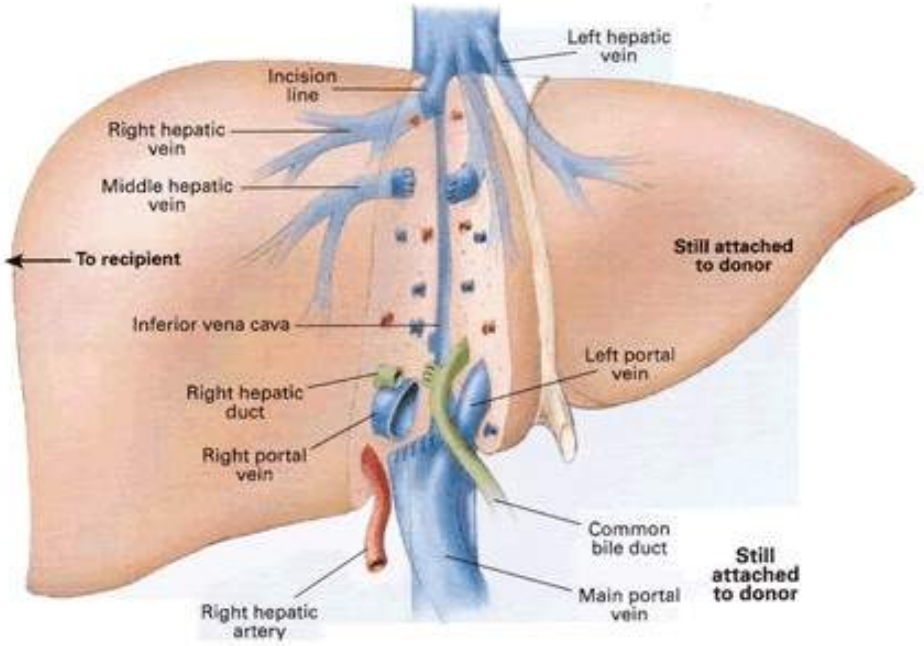
- **Bile Flow**



- Bile produced in hepatocytes→secreted into canalicul→bile ductules→common duct→gall bladder→bile duct→small intestine

Functions

- The liver has more than 200 functions, including:
  - Storage of Nutrients
  - Breakdown of erythrocytes
  - Bile Secretion
  - Synthesis of plasma Proteins
  - Synthesis of cholesterol

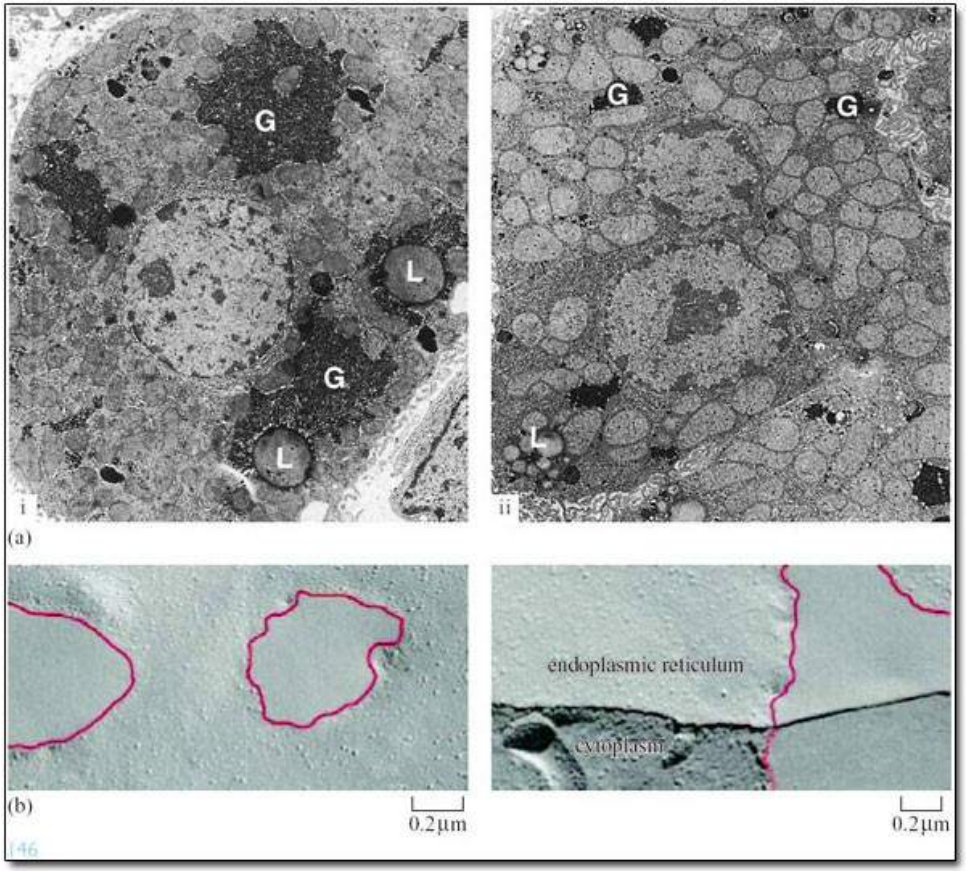


Trotter, Wachs, Everson, Kam; *Medical Progress: Adult-to-Adult Transplantation of the Right Hepatic Lobe.* The New England Journal of Medicine. April 4, 2002. Vol. 346. Page 1074-1082. Copyright 2002, Massachusetts Medical Society. All rights reserved.

Storage of Nutrients

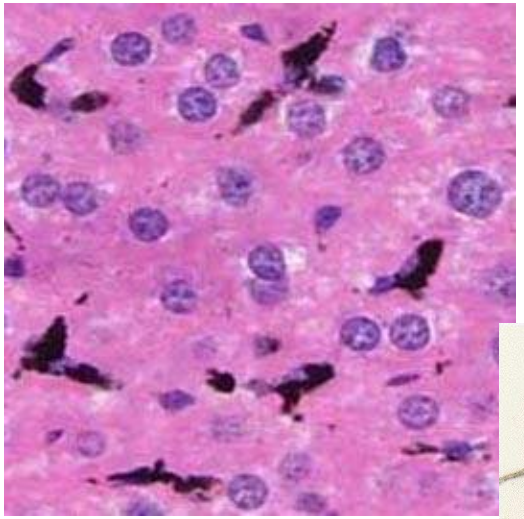
- Hepatocytes absorb and store excess nutrients in the blood
  - Glucose (glycogen)
  - Iron
  - Retinol (Vitamin A)
  - Calciferol (Vitamin D)

- Nutrients released when levels are too low



Breakdown of Erythrocytes

- RBC's have a life span of 120 days.
- RBC's weaken and rupturee, releasing hemoglobin into the blood plasma.
- Hemoglobin is absorbed by phagocytosis by Kuppfer cells in the liver.
- Hemoglobin is split into
  - Heme groups
    - Iron is removed from heme leaving a substance called bilirubin (bile pigment).
      - Iron is carried to bone marrow where it is used to new hemoglobin for RBC's
      - Bilirubin becomes a component of bile
  - Globins
    - Hydrolysed to amino acids and returned to the blood

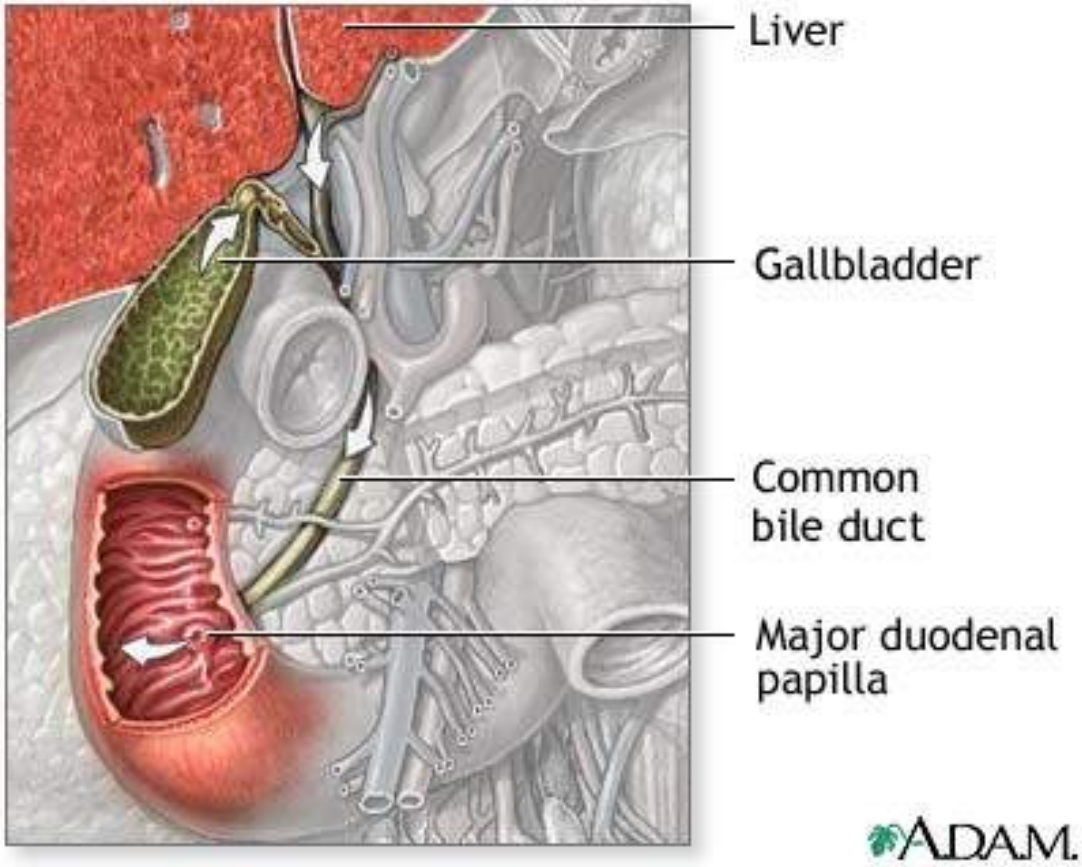




- An electron micrograph of a Kupffer cell from the liver. The Golgi apparatus (marked with arrows and *) is well developed. The dark granules associated with the Golgi saccules are lysosomes. At the cell surface, identify the filopodial processes.

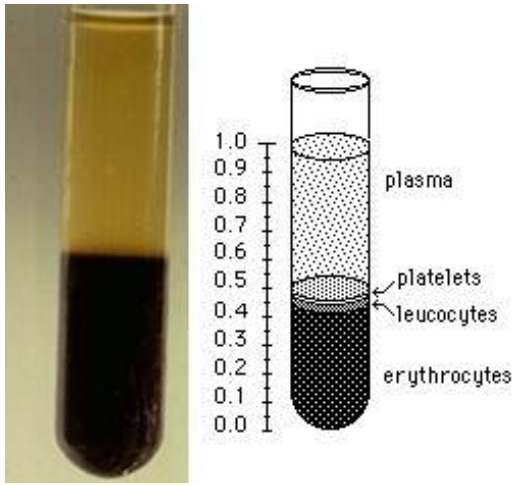
Bile Secretion

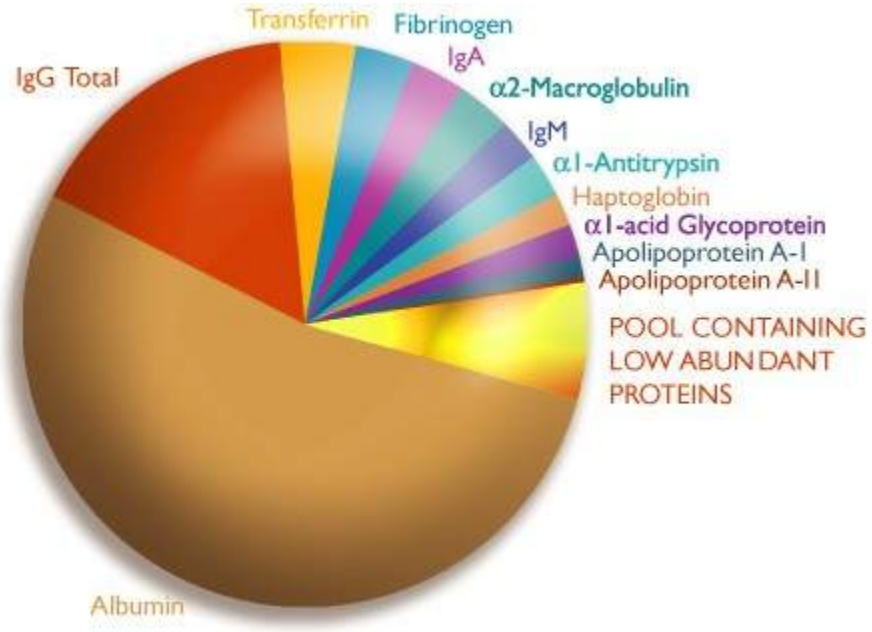
- Bile Contents
  - HCO_3^- (Bicarbonate)
  - Bile salts
  - Bile pigment
  - Cholesterol
- Stored in gall bladder
  - Concentrated
  - acidified
- Discharged into small intestine via bile duct



Synthesis of Plasma Proteins

- Produced by RER of Hepatocytes
- 3 main types
  - Albumin
  - Globulin
  - Fibrinogen





Synthesis of Cholesterol

- Produced by hepatocytes
- Some used for bile production
- Some trasnsported for use in the rest of the body
  - Synthesis and repair of cell membranes or stored in the liver.
  - Precursor by testis, ovaries or the adrenal gland to make steroid hormones.
    - progestins
    - glucocortoids
    - androgens
    - estrogens
    - mineralocortoids
  - It is also a precursor to vitamin D.

